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# Tool Use, Function Calling

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# From Text Generation to Action

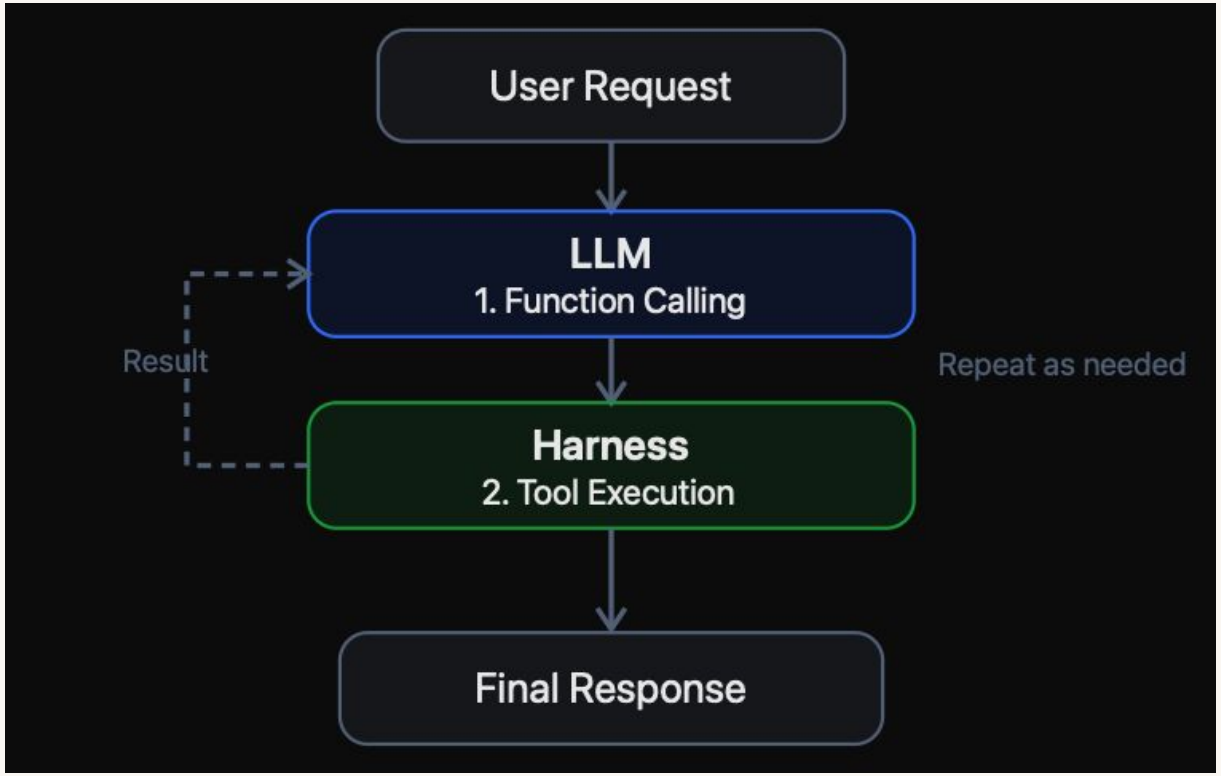
LLMs generate text. But what if we want them to perform tasks?

- LLMs take input and generate output.
- By themselves, they don't execute actions.
- We want LLMs to interact with the outside world.

**How do we turn an LLM that generates text into an agent that performs tasks?**

# How Tool Calling Works

- **Function calling**
  **The model decides which tool to call and with what arguments.**
- **Tool execution**
  **The harness runs the tool and returns its result.**
- **This process repeats until the model produces a final answer.**



# Defining a Function: Tool Schema

**How does the model know which tools are available?**

- **Tools are described using a JSON schema.**
- **Each definition specifies a name, description, and parameters.**
- **The schema tells the model what it can call.**

```json
{
  "type": "function",
  "name": "get_weather",
  "description": "Get weather for a city",
  "parameters": {
    "type": "object",
    "properties": {
      "city": { "type": "string" }
    },
    "required": ["city"],
    "additionalProperties": false
  }
}
```

# Function Calling: What the Model Does

**Models are trained to generate structured tool calls.**

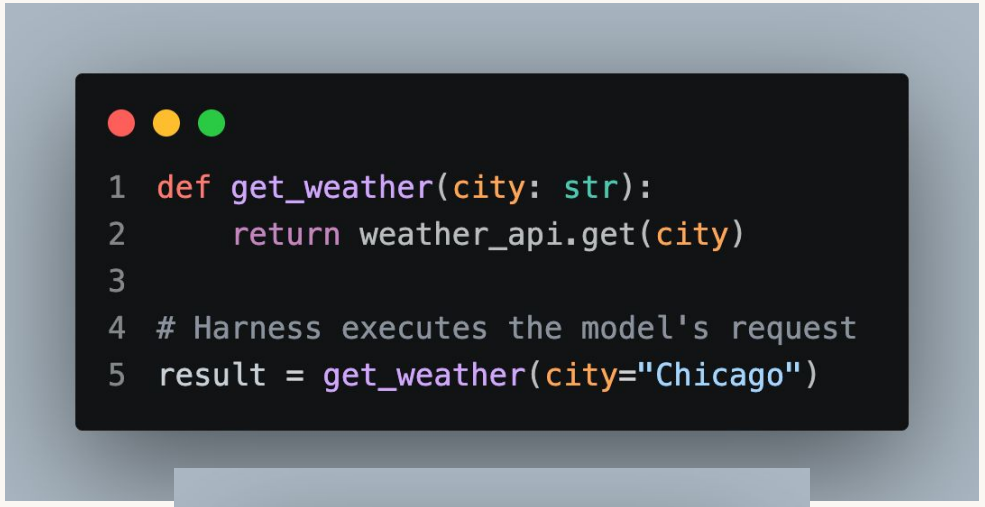
- **The model receives the user request and available tool schemas.**
- **It selects an appropriate tool.**
- **It generates arguments matching the schema.**
- **It does not execute the function itself.**

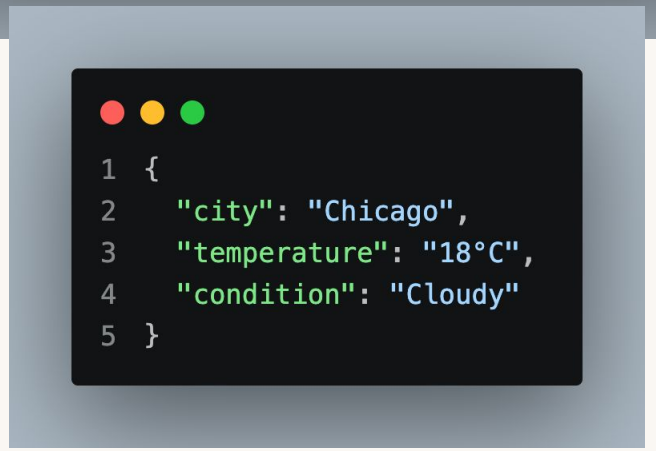


# Tool Execution: What the Harness Does

**The harness turns tool calls into real actions.**

- **Tools are implemented in code.**
- **The harness parses and validates the model's arguments.**
- **It invokes the matching tool.**
- **It returns the tool's output to the model.**



# Codex: A Real-World Agent

**Codex applies this architecture to software development.**

- **Common tools/capabilities:**

| Tool | Purpose |
|---|---|
| shell / exec_command | Run terminal commands, inspect files, execute tests |
| apply_patch | Create, modify, or delete code files |
| view_image | Inspect local images |
| update_plan | Track progress on multi-step tasks |

- **Codex uses these tools to interact with a codebase.**
- **Its harness manages execution, permissions, and results.**
- **Additional tools may be available depending on the environment.**


Demo



# THANK YOU